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Numerical study on molten pool dynamics and solute distribution in laser deep penetration welding of steel and aluminum

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ABSTRACT

To shed new light on the molten pool behavior and transmission mechanism during the laser welding of dissimilar metals, a 3D transient numerical model was developed in this study. The fluid flow, heat transfer, keyhole evolution, and mass transport between two dissimilar metals were studied. The volume of fluid (VOF) method was used to track the free surface. The effects of recoil pressure and surface tension on the keyhole wall were mainly considered. The effects of convection, diffusion, and keyhole formation on the mass transfer were also considered. The relative importance of convection and conduction and the effects of various driving forces on the convection in the molten pool were analyzed in terms of dimensionless numbers. The results showed that the maximum temperature and liquid flow velocity of the molten pool slightly fluctuate at the peak value after the molten pool enters the quasi-steady state, and the stability of the keyhole depends on the dynamic balance between the recoil pressure and the surface tension. Because of the influence of the formation mechanism of the keyhole, the weld zone contained significant amounts of aluminum elements. The results also showed that the welding parameters significantly affect the transition layer thickness and element distribution, which is closely related to the weld quality. The numerical simulation results correspond well with the experimental results. This study provides new insights into the weld pool dynamics in the laser welding of dissimilar metals and can help guide the selection of the welding parameters.

1. Introduction

Aluminum alloys, which can be characterized as lightweight high-strength materials, are widely used in the manufacturing of parts in the automobile and aerospace industries, with their usage being second only to steel [1,2]. High-strength steel is widely used in automobiles because of its corrosion resistance, high strength, low weight, and high elongation. Nowadays, considering the cost and crashworthiness, a hybrid steel–aluminum material is being used for car bodies; a steel–aluminum joining can help reduce the body weight and realize energy savings and emission reduction [3]. Lap welding is carried out with a steel plate on the top and aluminum plate on the bottom. This is done because of the low melting point of the aluminum plate and low reflectivity of the steel plate to the laser beam, thus saving energy consumption. Steel–aluminum joining technology is expected to be widely used in the body lightweight market. However, the joining of dissimilar materials is challenging in the structural design of hybrid bodies [4]. There are many difficulties in the joining of steel and aluminum materials. For example, the brittle compound FeAl is easily

formed at the welded joint, and a high thermal stress exists, leading to a poor quality of the welded joint and generation of cracks. Therefore, it is important to control the thickness of the intermetallic transition layer and heat input level. The thickness of the transition layer can be effectively controlled by fully penetrating the steel plate and partially penetrating the aluminum plate [5]. If the thickness of the intermetallic transition layer can be controlled in a reasonable range, the formation of the FeAl compound can be effectively reduced, and the quality of the welded joint can be improved [6]. This requires an accurate control of the heat input level per unit area of the workpiece. Compared with other joining methods, laser welding has advantages such as small heat affected zone, high energy density, high welding efficiency, deep penetration, and accurate and controllable welding process.

During laser welding, the energy transmission mechanism is influenced by the generation of the keyhole, and the laser beam directly acts on the workpiece through the keyhole. Pang et al. used a beam tracking model to simulate the evolution process of a weld pool-coupled keyhole surface, considering the multiple absorptions of the keyhole, and obtained a better weld morphology [7,8]. Cho et al. studied the stability of

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a keyhole in laser welding and established a 3D model considering heat and mass transfer. The results showed that the stable catenoid-like buttonhole formed in the welding process can improve the stability of the molten pool [9]. Pang and Chen simulated the dynamic change process of the molten pool and keyhole in laser welding, comprehensively considered the Fresnel absorption, heat convection, heat conduction, and evaporation effect of the phase interface, and found that the instability mechanism of the keyhole is closely related to the bulge at the keyhole wall [10,11]. Zhang et al. studied the behavior of dynamic keyhole in steel under high laser power [12]. Numerical and experimental results showed that the width of the keyhole reached a quasi-steady state much before the keyhole depth. The bulge of the front keyhole led to an excessive recoil pressure at the wall of the rear keyhole, leading to a serious deformation of the keyhole. An increase in the aspect ratio of the keyhole made it more unstable. Zhang used the ray tracing method and volume of fluid (VOF) method to simulate the laser welding of a full-penetration aluminum alloy, and analyzed the formation process of the porosity [13]. The research showed that under the condition of low surface tension, stagnation and solidification easily occurred at the root to form nodules, and the bulge on the front wall of the keyhole easily led to the collapse of the keyhole on the rear wall. These results are consistent with the conclusion drawn by Pang et al. [10,11] and Zhang et al. [12]. The above studies show that the formation of the keyhole has a significant influence on the flow and heat transfer of the molten pool and that the stability of the keyhole is closely related to the weld quality. Therefore, the effects of keyhole formation on the heat and mass transfer in the welding process of dissimilar metals are studied in this work.

Currently, scholars mainly focused on the weld joint morphology, mechanical properties, and temperature field in the laser welding of dissimilar metals; however, research on the keyhole behavior and solute transport mechanism in the molten pool is lacking. Yuce et al. studied the effect of laser heat input per unit area on the mechanical properties of welded joints of galvanized high-strength steel and aluminum alloy [14]. At relatively high heat input levels, a thick intermetallic compound was observed at the weld interface, which deteriorated the tensile strength of the joint. When the thickness of the intermetallic compound reaction layer was controlled within a certain range, a high-strength welded joint could be obtained. Huang studied the laser welding of steel and aluminum, obtained the temperature and stress distribution diagrams using the SYSWELD software, designed an orthogonal test, and effectively predicted the weld quality of a steel and aluminum joint using the response surface method [15]. Cao et al. studied the effect of Cu/Ni foil as an interlayer on the microstructure and properties of a laser-welded joint of steel and aluminum [16]. The addition of the Cu/Ni foil interlayer effectively improved the metallurgical reaction during laser welding and thereby the quality of the welded joint. Huang et al. established a 3D transient mathematical model of the keyhole instability in laser welding, and compared the formation mechanism of the porosity in the laser welding of a steel and aluminum alloy using the flow 3D software [17]. Although there is a significant difference in the porosities between steel and aluminum alloys, the flow trends in the molten pool are similar. Pan from Hunan University established a 2D transient model of dissimilar metals subjected to laser welding, analyzed the temperature and flow fields of the molten pool under a specific process, and further explored the influence of addition Ti powder on the FeAl compound distribution [18]. Given that only a 2D model was used, the formation mechanism of the keyhole at the cross section of the molten pool should be further studied.

In this work, a 3D transient numerical model of the laser welding of DP600 steel and 6082 aluminum alloy was established to analyze the dynamic behavior of the molten pool and solute distribution. A 3D conical Gaussian heat source was used in the numerical simulation, and the VOF method was used to track the gas-liquid free surface. The effects of recoil pressure, thermal buoyancy, surface tension, and Marangoni force on the flow of the molten pool were mainly considered. The

relative importance of the heat conduction and heat convection was compared in terms of the dimensionless number. The distributions of the steel and aluminum elements in the molten pool were studied. Finally, the effects of welding speed and laser power on the transition layer thickness were analyzed.

2. Numerical simulation

The laser welding of dissimilar metals involves various complicated physical phenomena, such as the fluid flow, heat transfer, mass transfer, and phase transformation. Because of the physical symmetry of the welding zone, only half of the welding zone can be selected as the calculation domain, and the length and width of the metal domain are reduced to improve the calculation efficiency. The thickness of the metal domain is the same as that of the welding sample. Fig. 1 shows the schematic of the laser welding of steel and aluminum. To simplify the mathematical model, we made the following assumptions:

- (1) The molten flow is laminar, and the molten metal is considered an incompressible and Newtonian fluid.
- (2) The heat source of the laser beam is modeled as a Gaussian density distribution.
- (3) The effect of shielding gas on the welding process is omitted.
- (4) In the model, only aluminum and iron elements are considered, whereas the other elements are ignored.
- (5) The mushy region is isotropic.
- (6) The gap between the two metal plates is ignored.

2.1. Governing equations

The simulation of the molten pool behavior in the laser welding of dissimilar metals is based on the mass, momentum, energy, and species transport, as expressed in Eqs. (1)–(4):

Mass equation:

$$\frac{\partial}{\partial t}(\rho) + \nabla \cdot (\rho \vec{V}) + S_{MA} = 0 \quad (1)$$

Momentum equation:

$$\frac{\partial}{\partial t}(\vec{V}) + \vec{V} \cdot \nabla \vec{V} = -\frac{1}{\rho} \nabla P + \mu \nabla^2 \vec{V} + S_{MT} + g \quad (2)$$

Energy equation:

$$\frac{\partial h}{\partial t} + \vec{V} \cdot \nabla h = k \nabla^2 T + S_E \quad (3)$$

Species transport equation:

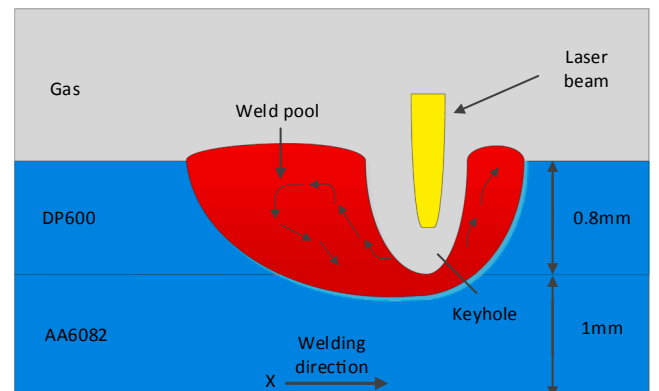


Fig. 1. Schematic of laser welding of steel and aluminum.

$$\frac{\partial}{\partial t}(\rho\gamma_i) + \nabla \cdot (\rho\bar{V}\gamma_i) = -\nabla \cdot \bar{J}_i \quad (4)$$

where $\bar{V}$, ρ , S_{MA} , p , g , μ , S_{MT} , h , k , T , S_E , and γ_i represent the velocity, density, mass source, pressure, gravity constant, dynamic viscosity, momentum source, enthalpy, coefficient of thermal conductivity, temperature, energy source, and mass fraction of species i , respectively. The mass source term S_{MA} in Eq. (1) mainly considers the gas-liquid mass conversion during evaporation and solidification. The momentum source term S_{MT} in Eq. (2) mainly includes the momentum loss during metal solidification, recoil pressure due to the rapid evaporation of the metal, and Boussinesq buoyancy due to the temperature gradient in the molten pool. The energy source term S_E in Eq. (3) mainly considers the laser heat source with a Gaussian distribution and the energy loss due to convection, radiation, and evaporation on the keyhole wall. $\bar{J}_i$ is the diffusion flux of species i . Since the melt is assumed to be laminar, $\bar{J}_i$ can be expressed as:

$$\bar{J}_i = -\rho D_{im} \nabla \gamma_i - D_{Ti} \frac{\nabla T}{T} \quad (5)$$

where D_{Ti} is the thermal diffusion coefficient, and D_{im} is the mass diffusion coefficient for species i in the mixture.

The VOF method was used to track the gas-liquid free surface. The equation of the volume fraction (F) can be described as [19]:

$$\frac{\partial F}{\partial t} + \bar{V} \cdot \nabla F = 0 \quad (6)$$

According to the definition, $F = 0$ when a cell has no liquid, whereas $F = 1$ when a cell is filled with liquid. When $0 < F < 1$, the liquid surface is in the cell.

The mass distribution of the different elements in the molten pool was calculated by solving the species transport Eq. (4). The dynamic viscosity μ , specific heat C_p , density ρ , and thermal conductivity k of the steel-aluminum mixture can be summarized as follows [20,21].

$$\mu = A\mu_1 + (1-A)\mu_2 \quad (7)$$

$$C_p = AC_{p1} + (1-A)C_{p2} \quad (8)$$

$$\rho = A\rho_1 + (1-A)\rho_2 \quad (9)$$

$$k = Ak_1 + (1-A)k_2 \quad (10)$$

Here, A is the mass fraction of the species. In the welding process, due to the change in the state between liquid and gas, the mass change can be expressed as:

$$S_{lg} = -m_{lg} \quad (11)$$

$$S_{gl} = m_{lg} \quad (12)$$

where m_{lg} is the cell mass, and S_{lg} and S_{gl} are the mass sources. The negative value represents the absorption of heat during liquid evaporation, and the positive value represents the release of heat during gas condensation.

2.2. Laser heat source model

The input of the laser heat source was treated as a body heat flux, and a 3D conical Gaussian heat source was adopted. The heat distribution radius of each section decreases linearly along the plate thickness direction [22].

$$q(r, z) = \frac{9\eta Q e^3}{\pi(e^3 - 1)H(t)(r_e^2 + r_e r_i + r_i^2)} \exp\left(-\frac{3r^2}{r_0(z)^2}\right) \quad (13)$$

and

$$r_0(z) = r_e - (r_e - r_i) \frac{z_e - z}{H(t)} \quad (14)$$

where η is the effective absorption coefficient of the laser, Q is the laser power, r_e and r_i are the effective radii of the heat sources on the top and bottom surfaces, respectively, where r_e is 0.0005 m and r_i is 0.0002 m. r is the distance from the workpiece to the central axis of the laser beam, $r_0(z)$ is the effective radius of the heat source in the z coordinate, and H is the depth of the heat source acts, which is a variable value and fluctuates with the phase interface.

In Eq. (13), the radius r can be expressed as:

$$r = \sqrt{(x - u_0 t)^2 + y^2} \quad (15)$$

where t is the welding time, and u_0 is the welding speed.

As mentioned previously, the heat source moves in the positive direction of the x -axis at the welding speed. The effective radius of the heat source decays along the thickness of the plate. In the laser welding process, the laser beam acts on the wall of the keyhole, and the defocus is set to zero.

2.3. Driving forces

The main driving forces of the molten pool are the recoil pressure, surface tension, buoyancy, gravity, and Marangoni stress. The recoil pressure is the driving force behind the formation of the keyhole, and the general expression is as follows [23]:

$$P_r = AB_0 / \sqrt{T_w} \exp\left(-\frac{U}{T_w}\right) \quad (16)$$

where P_r is the recoil pressure, U is a constant related to the material, T_w is the temperature of the keyhole wall, A is the adjustment coefficient, and B_0 is the evaporation constant. Since the recoil force P_r always acts vertically on the gas-liquid free surface, it can be decomposed into an axial component during calculation, and the component P_z along the z -axis is defined as follows:

$$P_z = \frac{n_z}{\sqrt{n_x^2 + n_y^2 + n_z^2}} \cdot P_r \quad (17)$$

where n_x , n_y , and n_z are the volume fraction gradient components along the x , y , and z axes, respectively. The other two axial recoil pressures P_x and P_y can be calculated using the same method.

The surface tension is expressed as in Eq. (18). T_m is the melting temperature of the workpiece, K is the curvature of the surface and γ is the surface tension coefficient of the liquid metal, which is calculated using Eq. (19).

$$P_\sigma = k\gamma \quad (18)$$

$$\gamma = \gamma_0 - \frac{d\gamma}{dT}(T - T_m) \quad (19)$$

The Boussinesq buoyancy due to the temperature difference can be expressed as:

$$S_b = \rho g \beta (T - T_m) \quad (20)$$

where B is the thermal expansion coefficient of the fluid.

2.4. Boundary conditions

As shown in Fig. 2, the top surface ABCD of the gas phase is set as the ambient temperature and velocity inlet, and the side surface of the gas phase is set as the ambient temperature and pressure outlet. The surface AEFB in the front end of the y -axis computational domain is set as the symmetric boundary. The outer surfaces of the workpiece

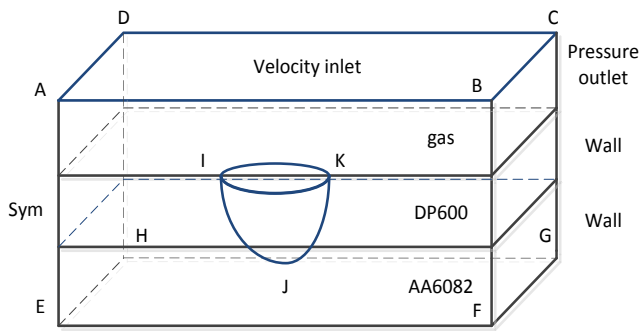


Fig. 2. Schematic of laser welding.

computational domains are set as wall boundaries [13], mainly considering the thermal convection and radiation. The wall boundary conditions are described as follows [24]:

$$k \frac{\partial T}{\partial n} = h_A(T - T_0) - \sigma \epsilon(T^4 - T_0^4) \quad (21)$$

where T and T_0 are the metal and ambient temperatures, respectively, σ is the Stefan–Boltzmann constant, ϵ is the surface radiation emissivity, and h_A is the convection coefficient.

For the gas–liquid free surface IJK, the energy balance mainly considers heat convection, heat radiation, and heat loss due to surface evaporation. The thermal boundary conditions can be expressed as [24]:

$$k \frac{\partial T}{\partial n} = q - h_A(T - T_0) - \sigma \epsilon(T^4 - T_0^4) - WH_{lg} \quad (22)$$

where q is the laser heat flux, W is the speed of evaporation, and H_{lg} is the latent heat of evaporation.

2.5. Numerical implementation

The finite volume method is used to discretize the mathematical model, and the commercial CFD software Fluent is used for the numerical simulation. The pressure-implicit with splitting of operator (PISO) pressure–velocity coupling method is used to solve the governing equations. The user-defined functions (UDFs) are implemented using the C programming language, which include the momentum source term (Eq. (16)), energy source terms (Eqs. (13) and (22)), boundary conditions, and initialization of the calculation domain.

Table 1 lists the chemical compositions of the DP600 high-strength steel and 6082-T6 aluminum alloy [14]. Tables 2 and 3 list the thermal physical properties and laser welding parameters, respectively. Based on the physical symmetry of the welding model, only half of the welding zone is used as the calculation domain to improve the calculation efficiency. The length, width, and height of the calculation domain are set to 8, 4, and 3 mm, respectively. Three subdomains were included: a gas phase with a height of 1.2 mm, steel with a height of 0.8 mm, and aluminum with a height of 1 mm. A hexahedral mesh is used, and the mesh size is set to 0.08 mm. The upper calculation domain is gas, the middle layer is steel, and the lower layer is the aluminum alloy. The calculation time step is set to 1×10^{-5} .

Table 1
Chemical compositions of DP600 and AA6082 (wt.%) [14].

	C	Mn	Si	Al	Cr	Mo	Mg	Fe
DP600	0.07	1.4	0.48	0.04	0.21	0.2	–	Base
AA6082	–	1	1.31	Base	0.25	–	1.2	0.5

Table 2
Physical properties of DP600 and AA6082.

Parameters	DP600	AA6082
Density	7800 (kg/m ³)	2700 (kg/m ³)
Dynamic viscosity	0.006 (kg/m·s)	0.0045 (kg/m·s)
Specific heat	475 (J/kg·K)	900 (J/kg·K)
Latent heat of fusion	2.47×10^5 (J/kg)	5.03×10^5 (J/kg)
Latent heat of evaporation	6.34×10^6 (J/kg)	1.07×10^7 (J/kg)
Thermal conductivity	29 (W/m·K)	90 (W/m·K)
Liquidus temperature	1727 (K)	889 (K)
Solidus temperature	1680 (K)	831 (K)
Evaporation temperature	3100 (K)	2790 (K)
Thermal expansion coefficient	4.4×10^{-5} (K ^{−1})	2.36×10^{-5} (K ^{−1})
Surface tension	1.2 (N/m)	0.99 (N/m)
Surface tension gradient	-4.9×10^{-4} (N/m·K)	-1.3×10^{-4} (N/m·K)
Radiation emissivity	0.4	0.4

Table 3
Simulation parameters of laser welding.

Power (W)	Welding speed (m/s)	Beam defocus (mm)	Heat input (J/mm)
1800	0.055	0	32.73
1900	0.055	0	34.55
2000	0.055	0	36.36
2100	0.055	0	38.18
2200	0.055	0	40.00
2000	0.045	0	44.44
2000	0.065	0	30.77
2000	0.075	0	26.67

3. Results and discussion

3.1. Model validation

Yuce et al. studied the laser lap welding of DP600 high-strength steel and 6082 aluminum alloy. Fig. 3(a) shows an image of the welding experiment conducted with a heat input of 36.3 J/mm [14]. Fig. 3(a) compares the experimental weld profile with the simulation results, in which the red solid line and green dotted line indicate the simulation and experimental results, respectively. Fig. 3(b) shows the cross-section morphology of the simulation results. The white dotted line is the dividing line between the two materials. The upper part is the steel plate, and the lower part is the aluminum plate. The experimental results show that the maximum penetration depth of the lower metal is 0.583 mm, the overall welding penetration is 1.383 mm, and the maximum weld width is approximately 1.4 mm. The simulation results show that the maximum penetration is 1.315 mm and that the maximum width is 1.381 mm. The shape and size of the simulated weld section are in good agreement with the experimental results. However, the upper surface of the weld is slightly different, which may be due to the lack of consideration of the zinc coating and some assumptions made for the simulation.

3.2. Heat transfer and fluid flow

Fig. 4 shows the calculated 3D temperature and velocity field for the Case with $P = 2000$ W and $V = 0.055$ m/s. At the beginning, the laser beam acts on the surface of the workpiece, the temperature of the heating zone increases, and the metal begins to melt and form a molten pool, as shown in Fig. 4(a). As the welding progresses, the weld pool size gradually increases, and the heat affected zone expands gradually. The laser beam moves along the x-axis at a speed of 0.055 m/s, and the molten metal behind the laser beam reconsolidates to form a weld. When $t = 18.6$ ms, the molten pool reaches a quasi-steady state, and a typical V-shaped profile is formed at the tail because of the latent heat of phase transition, which is consistent with the results reported by Geng [25].

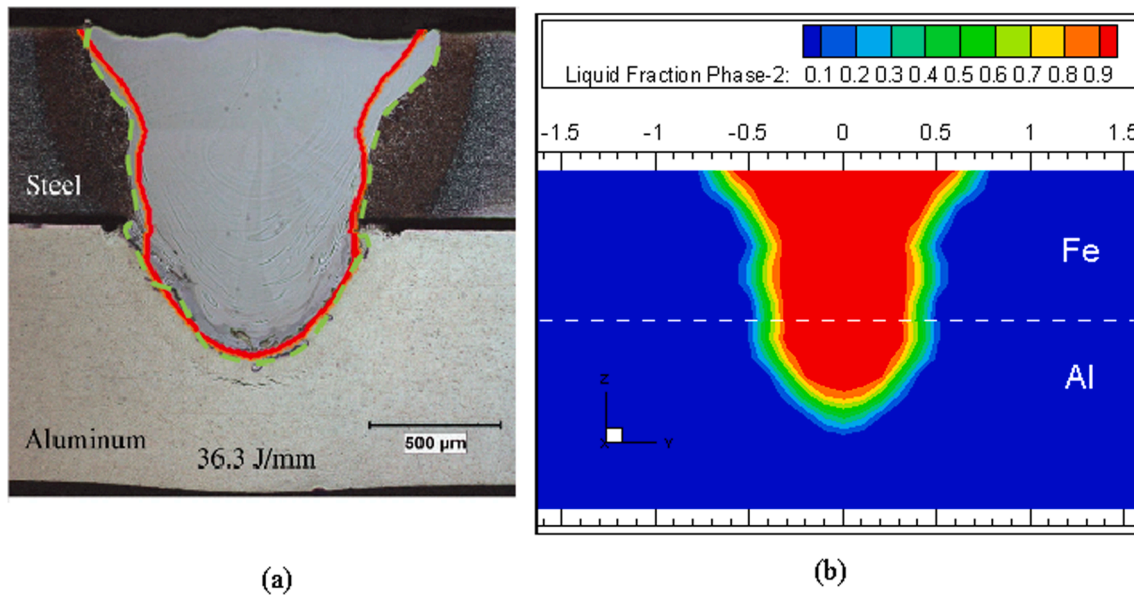


Fig. 3. Cross-sectional morphology of the molten pool: (a) Cross-section of the welding experiment [14] with the fusion lines (red real line: simulation, green dotted line: experiment), (b) Cross-section of the simulation.

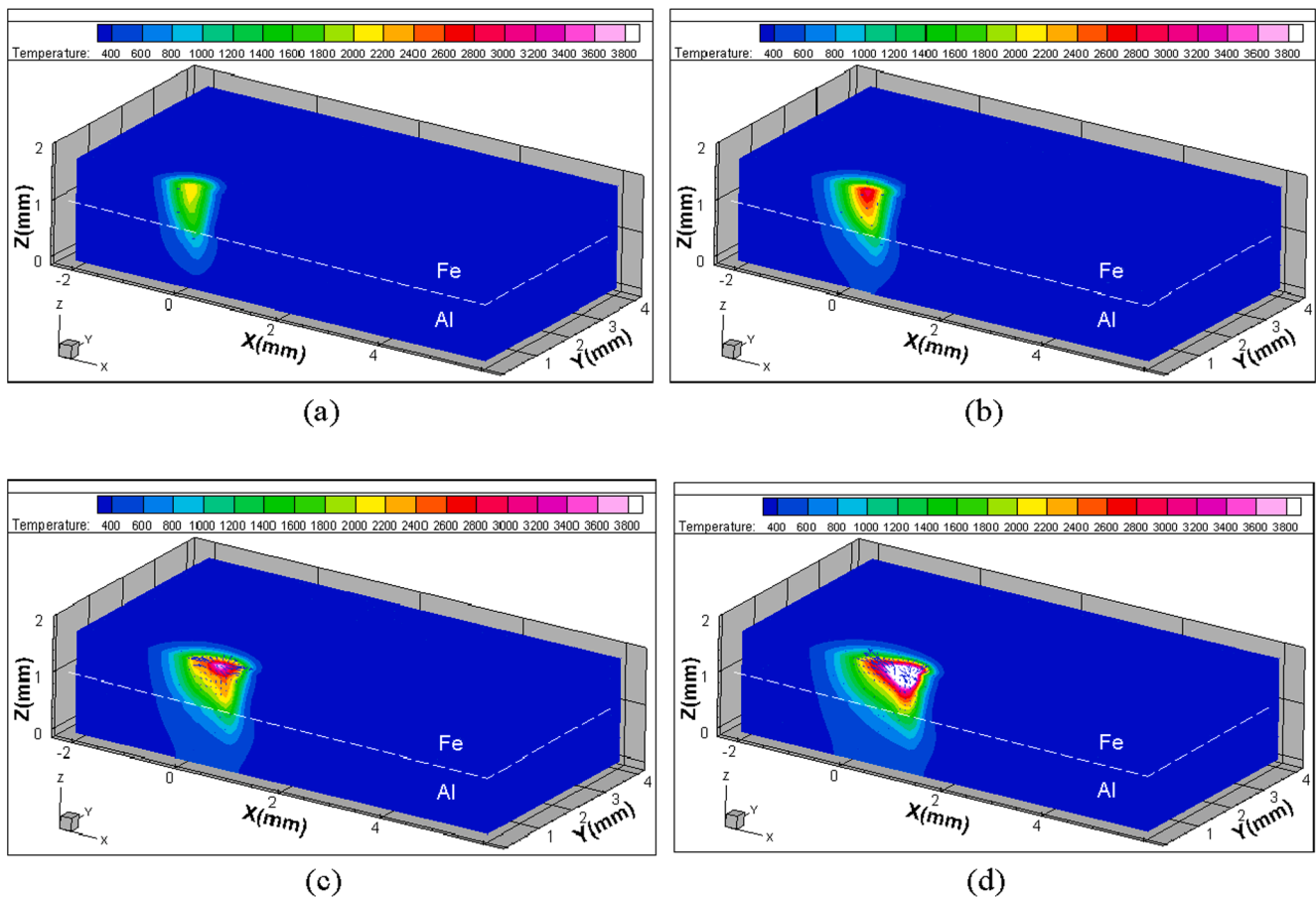


Fig. 4. Distributions of the temperature and fluid velocity at different times: (a) $t = 5$ ms, (b) $t = 10$ ms, (c) $t = 14$ ms, (d) $t = 18.6$ ms.

From Fig. 4(c), a high-temperature region is formed inside the workpiece, and the maximum temperature reaches 3800 K, exceeding the boiling point of the material. Under the action of recoil pressure, the metal near the keyhole tends to flow upward. The annular convection is formed at the back of the keyhole. The temperature near the keyhole is

evidently higher than that in the mushy area, forming a larger temperature gradient. The surface tension of the molten metal decreases significantly with the increase in the temperature, and the surface tension coefficient near the keyhole is negative. From the top view of the workpiece shown in Fig. 5, it is evident that the metal liquid flows

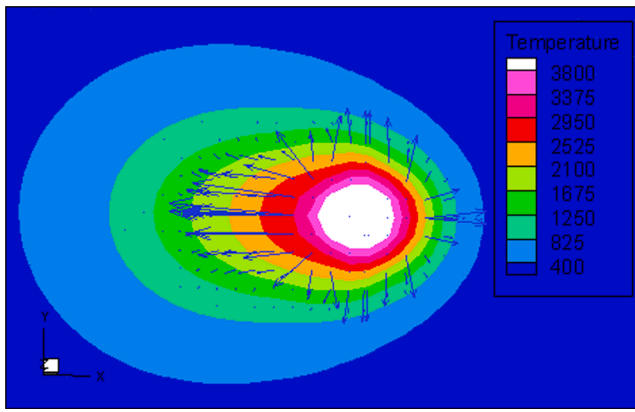


Fig. 5. Top view of temperature and velocity distributions.

outward from the center of the keyhole under the effect of the negative surface tension coefficient. The Marangoni force is due to the surface tension gradient. A strong Marangoni convection effect can be observed near the keyhole with a large temperature gradient, and the maximum flow velocity is 30.1 m/s.

Fig. 6 shows the distributions of the temperature and fluid velocity at the cross section of the workpiece at different times; the distributions are symmetrical. With the continuous development of welding, the maximum temperature of the molten pool increases, and the liquid flow velocity increases significantly until the molten pool enters the quasi-steady state.

After entering the quasi-steady state, the maximum flow velocity and maximum temperature of the molten pool fluctuate slightly. Because the laser beam is moving, the curvature of the keyhole surface varies, resulting in a change in the recoil pressure and surface tension. Furthermore, because of the influence of plasma formed in the keyhole, the heat input and output oscillate slightly. Therefore, the dynamic balance between the driving force and the energy leads to fluctuations in

the flow and temperature, and the reasons for the same are described in detail in the next section.

3.3. Evolution of keyhole

Fig. 7 shows the keyhole evolution process in the laser deep penetration welding of the dissimilar metals under the conditions of $P = 2000$ W and $v = 0.055$ m/s. The red area in Fig. 7 represents the metal calculation domain, the upper part of the dotted line represents steel, and the lower part represents the aluminum alloy. In the initial stages of welding, the metal workpiece absorbs heat under the action of the laser beam and partially melts. When the temperature of the workpiece gradually rises to the boiling point, a small amount of liquid metal begins to evaporate, as shown in Fig. 7(a). With the continuous development of welding, the liquid metal starts flowing in large quantities and evaporate violently, producing a significant amount of plasma. With the increase in the recoil pressure and the decrease in the surface tension, as shown in Fig. 7(b), more and more liquid metals are pushed out of the concave surface to form keyholes under the action of the recoil pressure. After the keyhole is formed, the heat transfer mechanism changes from heat conduction welding to deep penetration welding, and the laser beam acts on the keyhole wall directly. Heat is transferred to the workpiece along the keyhole, and the depth and width of the keyhole increase gradually. With the increase in the welding time, the keyhole enters the quasi-steady state, and the depth of the keyhole tends to stabilize but oscillates slightly near the steel–aluminum interface.

After the molten pool enters the quasi-steady state, the surface tension and recoil pressure reach a dynamic equilibrium. When the plasma gathers in the keyhole, it can shield and reflect the laser to a certain extent, the heat absorbed by the workpiece decreases, and the temperature decreases, which decreases the recoil pressure at the keyhole. When the recoil pressure is less than the surface tension, the keyhole tends to shrink, and the volume decreases. When the temperature decreases, the amount of plasma also decreases, the shielding and reflection of the laser by the plasma are weakened, and the heat absorbed by

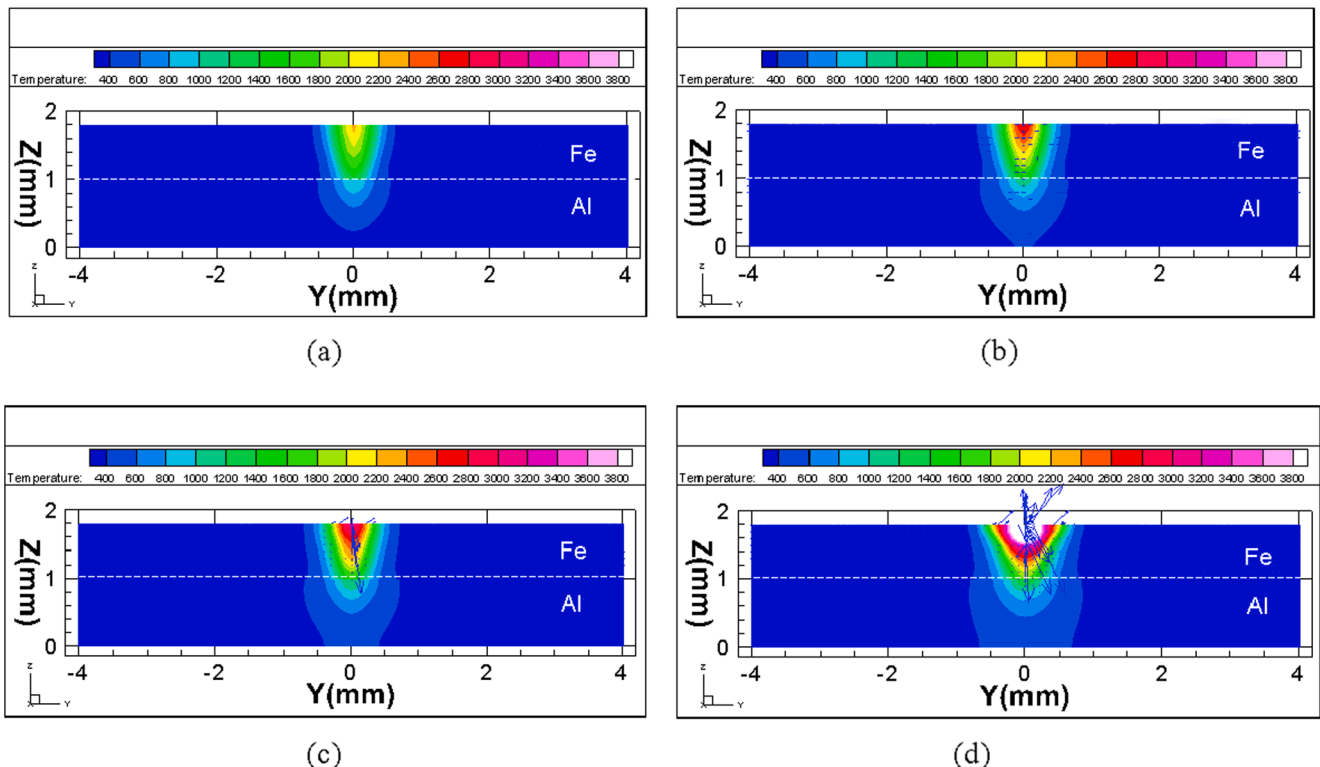


Fig. 6. Distributions of fluid velocity and temperature at the YZ cross-section at different times: (a) $t = 5$ ms, (b) $t = 10$ ms, (c) $t = 14$ ms, (d) $t = 18.6$ ms.

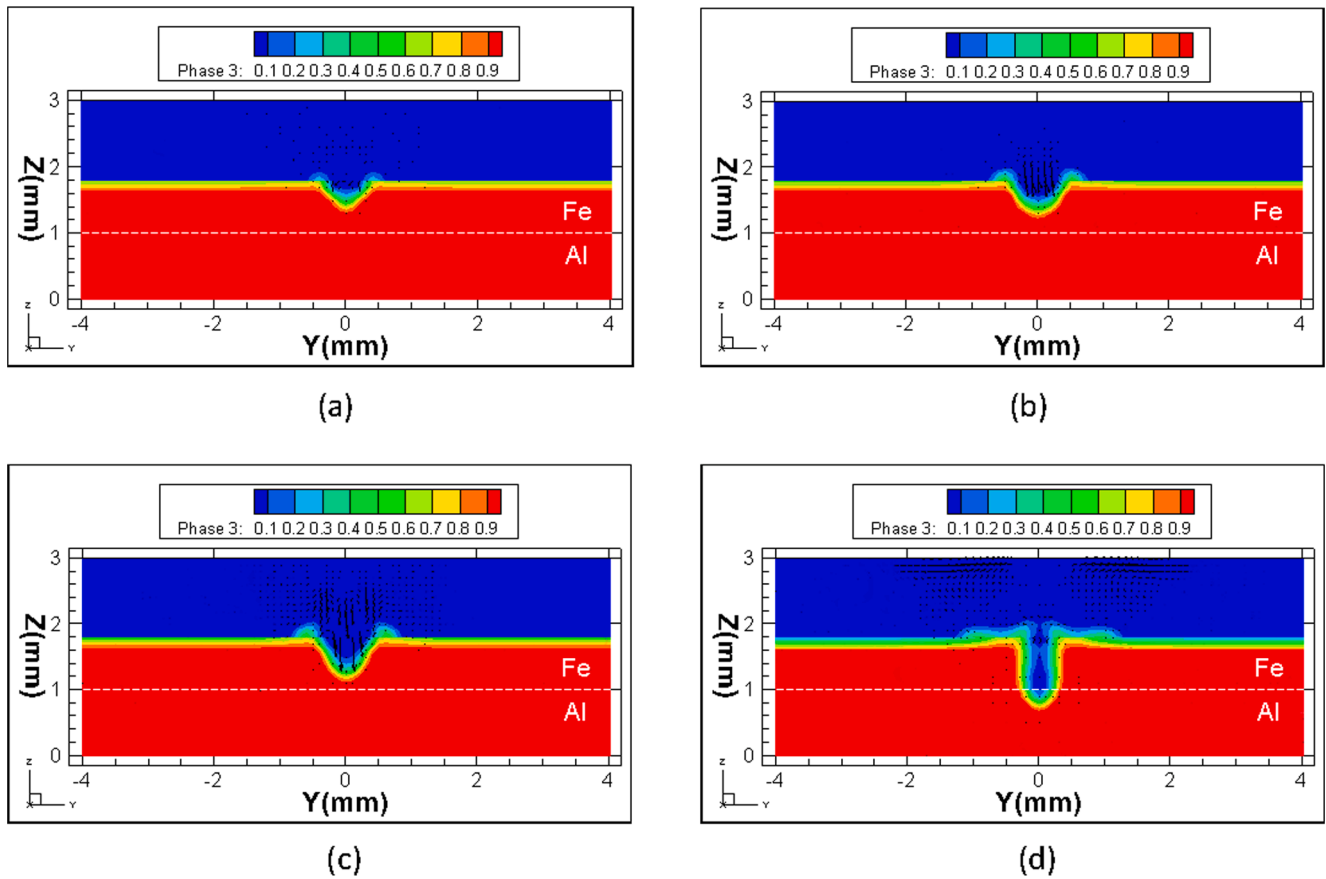


Fig. 7. Evolution of the keyhole at the YZ cross-section of laser welding.

the workpiece increases, leading to a rise in the temperature. When the recoil pressure is greater than the surface tension, the keyhole expands again. Therefore, the keyhole will be in a state of periodic fluctuation. The stability of the keyhole is closely related to the dynamic balance between the recoil pressure and the surface tension. The change in the driving force leads to a fluctuation in the flow velocity, and the change in the laser absorption effect explains the fluctuation in the temperature. The iso-surface in Fig. 8 represents the gas-liquid interface; we find that the upper surface of the keyhole is not smooth, and there is a certain bulge. This is due to the incomplete evaporation of the liquid metal after being pushed out of the surface under the action of the recoil pressure. This explains the unevenness of the weld surface. Furthermore, it can be found that there is a large upward velocity vector in the keyhole. This shows that there is spatter phenomenon in the hole, which is consistent with the conclusion drawn by Zhang et al. [13].

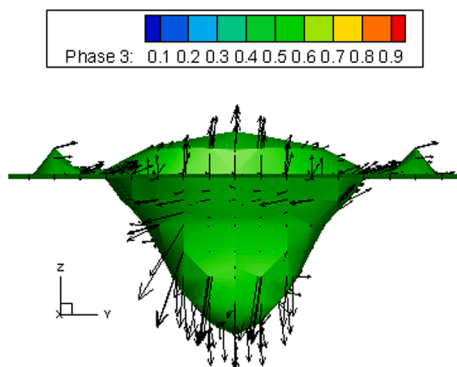


Fig. 8. Iso-surface of the keyhole.

In the process of dissimilar metal welding, the heat input level per unit area must be controlled accurately. If the heat input per unit area is too high, a thick metal reaction layer will be formed, thus easily leading to the formation of intermetallic compounds at high temperatures [20]. Intermetallic compounds have the characteristics of high brittleness and high hardness. They are the weakest location in the joining and are prone to brittle fracture. In comparison, a low heat input level per unit area will result in a low temperature at the aluminum plate, insufficient melting, and ineffective joining. Therefore, an accurate control of the heat input level is of great significance for improving the weld quality. As shown in Fig. 7(d), the steel of the upper layer is penetrated fully, whereas the aluminum of the lower layer is penetrated locally, and the reaction layer is controlled within a certain range. The above method helps ensure an effective joining between the materials, reduce the formation of intermetallic compounds, and improve the mechanical properties of the weld.

3.4. Role of dimensionless number in molten pool

The heat transfer in the molten pool is mainly realized by convection and conduction. The Peclet number is used to evaluate the relative importance of the heat convection and heat conduction in the molten pool. The Peclet number is derived from the ratio of the convective heat transfer to the conductive heat transfer:

$$Pe = \frac{\text{heat}_{\text{convection}}}{\text{heat}_{\text{conduction}}} = \frac{\rho C_p u L_R}{k} \quad (23)$$

where u is the characteristic velocity, and L_R is the characteristic length. The cross section of the molten pool is considered a rectangle, as shown in Fig. 9. The depth of the molten pool h is taken as the rectangle height, and the average half width $\bar{w}$ of the molten pool is taken as the rectangle

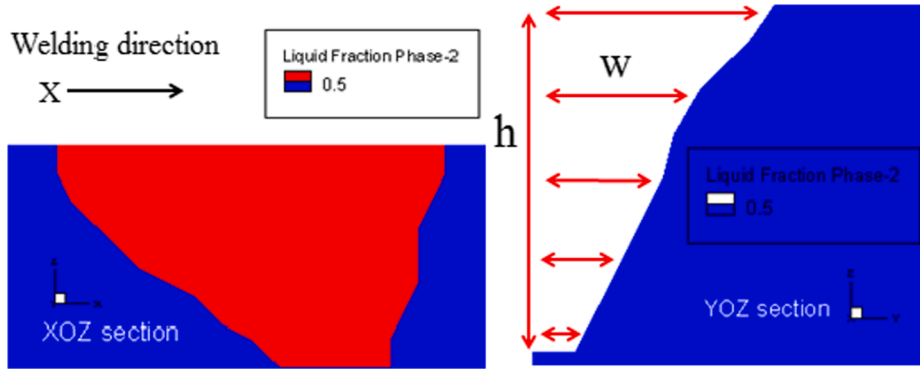


Fig. 9. Definition and measurement diagram of the characteristic length.

length. Here, $\bar{w}$ is the average value of the half width at five different heights. The characteristic length can be expressed as:

$$L_R = \frac{4S}{C} = \frac{2\bar{w}h}{\bar{w} + h} \quad (24)$$

where C and S are the wetting perimeter and cross-sectional area of the flow channel, respectively. The characteristic length and velocity vary with the welding time; Fig. 10 shows the variation in the dimensionless number with time. When Pe is much greater than 1, the convective heat transfer is dominant. When Pe is less than 1, the main mechanism of heat transfer is heat conduction. At the beginning, the Peclet number is approximately 10. With the increase in the weld pool dimension, the Pe number increases, and the convection becomes more intense. After the molten pool enters the quasi-steady state, the Pe number becomes stable at approximately 80. Therefore, in the laser deep penetration welding of steel and aluminum, the convective heat transfer is always dominant.

The influences of different forces on the molten pool flow can be expressed in terms of the following dimensionless number. Here, the ratio of the surface tension to the viscous force can be expressed as:

$$Ma = \frac{L_R \rho \Delta T |\partial \gamma / \partial T|}{\mu^2} \quad (25)$$

where ΔT is the difference between the maximum temperature in the liquid metal and the solidus temperature, and $|\partial \gamma / \partial T|$ is the temperature gradient. The ratio of the buoyancy to the viscous force can be expressed as:

$$Gr = \frac{L_R^3 g \beta \Delta T \rho^2}{\mu^2} \quad (26)$$

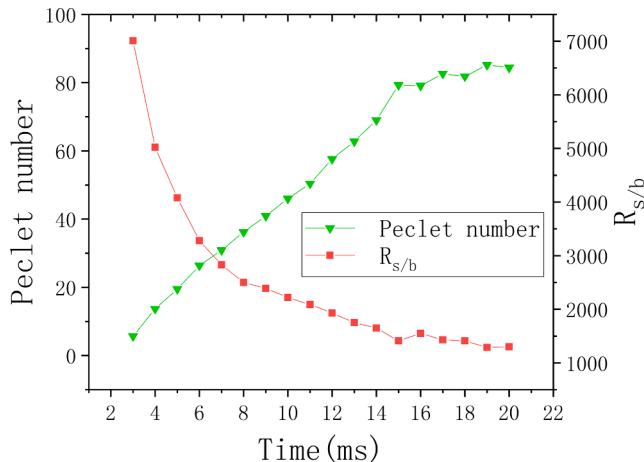


Fig. 10. Dimensionless number at different times.

The ratio of the surface tension to the buoyancy force is calculated as:

$$R_{s/b} = \frac{Ma}{Gr} = \frac{|\partial \gamma / \partial T|}{g \beta \rho L_R^2} \quad (27)$$

From Fig. 10, the $R_{s/b}$ number decreases with the increase in the dimension of the molten pool, whereas the $R_{s/b}$ number is still much greater than 1. Therefore, the fluid flow is mainly driven by Marangoni convection.

3.5. Mass transfer

Convection and diffusion are two important forms of mass transfer. As there is no convection in a solid, diffusion is the only mass transport mechanism. The diffusion studied in this work follows the Fick's law, i. e., matter always diffuses from a high concentration to a low concentration. In the numerical calculation, the element concentration distribution was obtained by solving the mass transport equation, i. e., Equation (4). As shown in Fig. 11, a "pit"-shaped transition layer is formed at the welding joint of the steel and aluminum. The aluminum element in the lower layer enters the weld zone by convection and diffusion, and its concentration is 11.7 at% as measured using the probe method in the Tecplot software. This shows that a large amount of aluminum enters the weld zone and that the element concentration is relatively uniform, as shown in the figure. The transition layer is the area where intermetallic compounds are most easily formed, in which the concentrations of the steel and aluminum are very high. The aluminum concentration increases gradually from the weld zone to the transition layer and is maximum after leaving the transition layer. The distribution of the steel element concentration is different from that of the aluminum element concentration. The concentration of the steel elements in the weld zone and transition layer is high. However, below the transition layer, the concentration decreases sharply and is basically zero. This is mainly due to the solid area below the transition layer, and iron can only enter a lower metal domain by diffusion. However, in the weld zone, the molten pool flows violently, and the aluminum element enters the weld zone by convection and diffusion. Therefore, the concentration of the aluminum through the transition layer is higher than that of steel, indicating that convection plays an important role in metal mass transfer. An enlarged figure of the transition layer is shown in Fig. 11(b), and the steel concentration increases gradually along the AB direction. Fig. 12 shows the variation in the mass fraction of the steel along the Z axis. Fig. 11(c) shows the phase distribution diagram at this time. The phase diagram shows that the small hole just penetrates the lower metal domain.

Fig. 13 shows the distribution of the steel concentration at the YOZ section calculated under a laser power of 2000 W and a welding speed of 0.055 m/s. In this paper, the transition layer is defined when the concentration range of steel and aluminum is between 30% and 70%. When the welding speed is maintained at 0.055 m/s and the laser power is

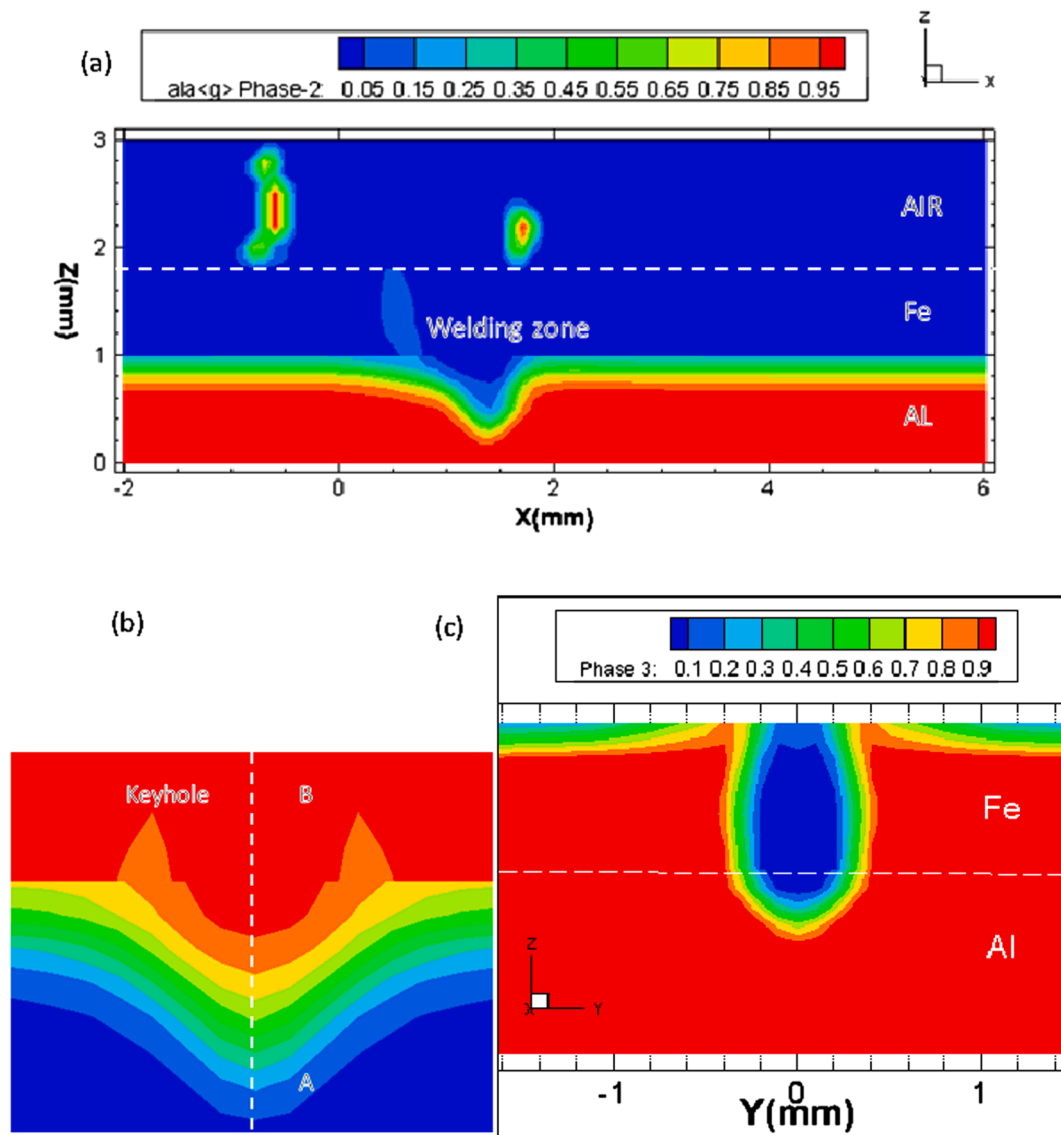


Fig. 11. Transition layer of steel–aluminum welded joint: (a) Distribution diagram of aluminum concentration, (b) Enlarged drawing of the transition layer, (c) Phase distribution diagram.

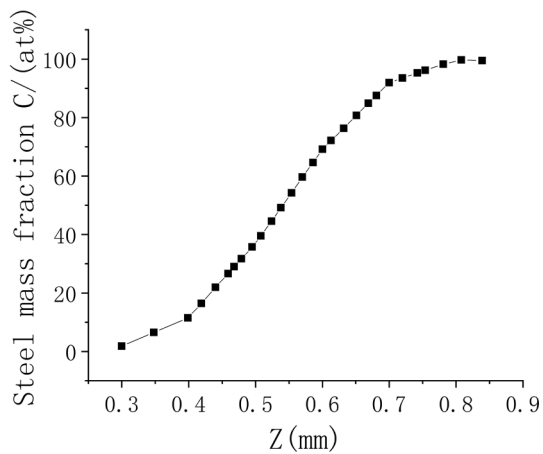


Fig. 12. Distribution of the steel mass fraction along the Z coordinate.

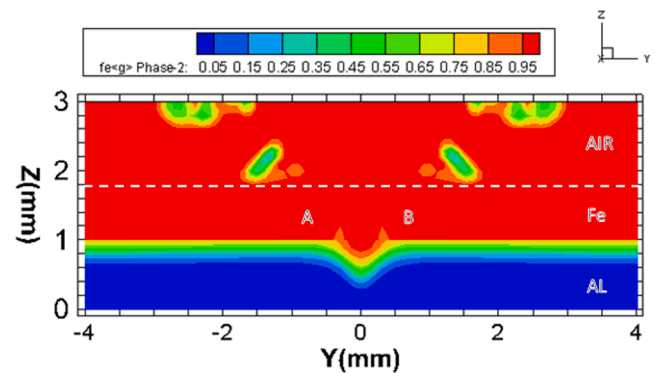


Fig. 13. Distribution of the steel concentration at the YOZ section.

increased from 1800 W to 2200 W, the thickness of the steel–aluminum laser welding transition layer varies with the laser welding power, as shown in Fig. 14. As shown, with the increase in the laser power, the transition layer thickens gradually; moreover, the heat input per unit

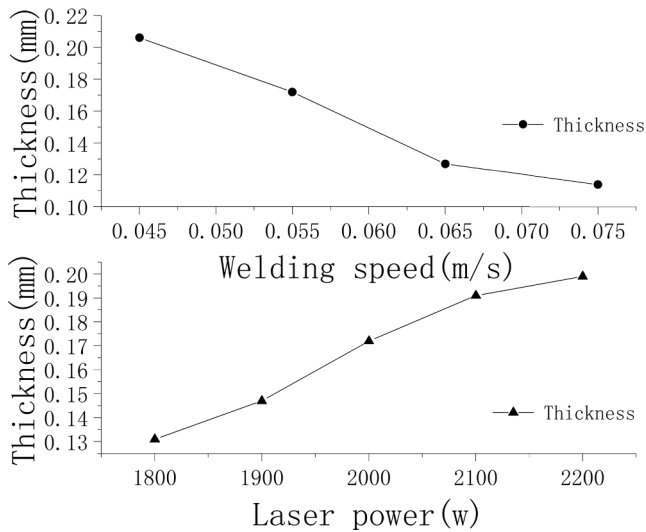


Fig. 14. Variation in the transition layer thickness with welding process parameters.

area increases, the depth of the keyhole increases, and the flow of the molten pool becomes more intense. Therefore, the effect of metal mass transfer is enhanced, and the thickness of the metal transition layer is also increased. When the laser power is maintained at 2000 W, and the welding speed is increased from 0.045 m/s to 0.075 m/s, the transition layer thickness varies with the welding speed, as shown in Fig. 14. The transition layer thickness decreases with the increase in the welding speed. When the laser welding speed increases, the high temperature residence time of the weld pool is shortened, thereby decreasing the keyhole depth and flow velocity of the molten pool.

In addition to convection and diffusion, keyhole formation is an important factor in the metal mass transport during welding. From Fig. 13, the aluminum element concentration in regions A and B near the keyhole is higher, which is closely related to the keyhole formation. According to the evaporation mechanism, the molten metal evaporates violently under the action of the laser beam, and the recoil pressure acts on the wall of the keyhole. When the keyhole penetrates the lower metal, the lower metal aluminum is squeezed into the upper metal domain under the action of the recoil pressure. Therefore, the aluminum concentration in regions A and B is higher. Furthermore, regions A and B are prone to form intermetallic compounds at high temperatures, leading to welding defects. Therefore, it is necessary to control the depth of the keyhole.

4. Conclusions and prospect

In this work, the laser welding of AA6082 aluminum alloy and DP600 steel under different parameters were studied. A 3D numerical model was established to study the keyhole behavior and weld pool dynamics. Furthermore, the mass transport between the steel and aluminum was calculated. The major conclusions of the study are as follows:

- (1) When the laser power was 2000 W, and the welding speed was 0.55 m/s, the maximum liquid velocity and temperature of the molten pool were 30.1 m/s and 3800 K, respectively. Subsequently, the temperature and liquid velocity slightly fluctuated at the peak value.
- (2) The convective heat transfer was found to be dominant in terms of the dimensionless Peclet number. The fluid flow was mainly driven by surface tension. The stability of the keyhole depended on the dynamic balance between the recoil pressure and the surface tension.

- (3) Convection and diffusion are the main mechanisms of metal mass transport, and different welding processes have significant effects on the metal mass transfer. When the heat input per unit area decreased, the fluid flow, element diffusion, and thickness of the intermetallic transition layer decreased.
- (4) The formation of the keyhole will make the metal in the lower region to squeeze into the upper layer, which has an important influence on the distribution of the element concentration.

The influence of the shielding gas was ignored in this work, because the laser welding process is very complicated. To improve the accuracy of the model, the influence of shielding gas on welding simulation results can be considered in the future. Based on this work, the influence mechanism of the molten pool behavior on the solidification structure can be further explored. The heat and mass transfer mechanism in dual-beam laser welding can also be studied.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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